HOTEL MANAGEMENT SYSTEM

SRS REPORT

Submitted By:

Ishfak Akbar Nahian

ID: 232-134-028

Mahir Ihsan Chowdhury

ID: 231-134-036

SWE 5th

Submitted To:

Wadia Iqbal Chowdhury

Senior Lecturer

Department of Software Engineering

TABLE OF CONTENTS

ABSTRACT

The Hotel Management System (HMS) is a robust and user-friendly software designed to automate and streamline the key operational processes within a hotel. It serves as a centralized platform for managing critical hotel functions such as room reservations, guest check-ins and check-outs, customer data management, billing, and financial calculations. By offering features such as customer tracking, automated tax calculations, and intuitive data handling, the system significantly reduces manual effort, minimizes errors, and enhances overall efficiency.

The HMS interface is designed with simplicity and usability in mind, ensuring that hotel staff can quickly adapt to the system without extensive training. It includes essential features like the ability to add new records, search for existing records, update customer information, calculate total costs (including taxes), and generate clear billing summaries. The system also supports the categorization of rooms, customer nationality selection, and meal plan integration, providing flexibility to accommodate the unique needs of each guest.

In addition to operational efficiency, the HMS emphasizes data accuracy and security, ensuring that all customer and transaction information is stored systematically and can be retrieved quickly for reporting and administrative purposes. With this system, hotels can enhance the guest experience by reducing wait times during check-in/check-out, providing personalized services, and maintaining consistent and reliable operations. The HMS is a scalable solution suitable for small to medium-sized hotels aiming to modernize their workflows and deliver exceptional service.

OBJECTIVE

- **Simplify Customer Management:** The system captures key guest details (e.g., name, contact, ID, and nationality) through well-structured input fields, enabling quick and efficient record handling.
- **Automate Billing and Payments:** Fields like "No of Days," "Subtotal," and "Total Cost" automate financial calculations based on check-in/out dates, reducing errors and saving time.
- **Efficient Room Allocation:** Features like "Room Type," "Room No," and "Room Ext. No." help manage room assignments and availability to optimize resource utilization.
- **Ensure Data Accessibility:** Guest records are displayed in a detailed list format for quick access, while the search function simplifies locating specific entries.
- **User-Friendly Interface:** Buttons like **"AddNew/Total", "Search"** and **"Reset"** make key operations straightforward, ensuring ease of use for staff with minimal training.
- **Enhance Guest Experience:** Automated processes reduce delays in check-in/check-out, while features like meal plans and room customization provide personalized services.
- **Scalable and Reliable Operations:** The system can handle multiple records seamlessly, ensuring reliability and scalability for growing hotel operations.

INTRODUCTION

The Hotel Management System (HMS) is designed to make hotel operations easier by automating key tasks such as managing customer information, assigning rooms, processing payments, and handling check-ins and check-outs. The system's user-friendly interface makes it easy for hotel staff to use, improving efficiency and reducing errors.

It automates room assignments, ensuring rooms are allocated correctly and without delays. The system also manages billing, automatically calculating charges and taxes, making the checkout process smoother for both guests and staff.

For check-ins and check-outs, the HMS simplifies the process, reducing wait times and keeping records up-to-date in real time. This ensures that staff have quick access to important guest details.

Designed for small to medium-sized hotels, the HMS helps improve hotel operations, save time, and enhance the overall guest experience by making everything run more smoothly and efficiently.

PURPOSE

The main goal of the Hotel Management System (HMS) is to make hotel management easier and more efficient by automating routine tasks and reducing mistakes. It provides a single platform for managing customer records, room assignments, and financial transactions, which helps improve how the hotel operates.

By automating tasks like room bookings, billing, and check-ins, the system ensures that everything is accurate and up-to-date. This reduces errors, such as double bookings or incorrect charges, and speeds up processes.

The system also makes it easier for staff to access customer information and update records in real time, helping the hotel run smoothly. This leads to faster service and better coordination between departments.

In the end, the HMS improves both operational efficiency and the guest experience, ensuring guests receive accurate billing, quicker check-ins, and more personalized service, while staff can manage tasks more effectively.

SCOPE

The HMS is designed to support essential hotel operations, including:

1. Capturing and storing customer details such as name, contact information, and ID proofs.
2. Managing room bookings, availability, and assignments based on room type and preferences.
3. Automating billing, tax calculations, and generating final invoices.
4. Displaying and searching guest records in an organized format for quick access.
5. Providing a scalable solution that can adapt to growing operational demands.

This system is suitable for small to medium-sized hotels and can potentially be extended to include features like loyalty programs or integration with other services (e.g., restaurants).

OVERVIEW

The Hotel Management System (HMS) provides a simple and efficient way to manage hotel operations. It allows staff to:

- Add, update, and search for customer records.
- Handle room bookings, assignments, and availability.
- Automate billing with tax calculations and generate final bills.
- Organize data in an easy-to-access format for quick updates and reports.

With its intuitive design and essential features, the system makes daily tasks easier, reduces errors, and enhances the overall guest experience.

SDLC MODEL

WHICH MODEL CAN BE USED?

The Software Development Life Cycle (SDLC) is a process model used to design, develop, test and maintain a software application.

Various Software Development Life Cycle (SDLC) models can be applied to the Hotel Management System (HMS) based on the project's scope, requirements, and limitations. The following is an evaluation of suitable models based on the provided details:

ADVANTAGES AND DISADVANTAGES:

1. Waterfall Model:

- **Advantages:**
  - Clear, structured approach with defined phases.
  - Easy to understand and manage for straightforward projects with stable requirements.
  - Well-suited for projects like HMS, where the requirements are clearly outlined at the start.
- **Disadvantages:**
  - Inflexible to changes in requirements once the development has started.
  - Late testing phase may lead to costly changes if major issues are found.

2. Agile Model:

- **Advantages:**
  - Flexible to changing requirements and client feedback.
  - Promotes iterative development with frequent releases and customer involvement.

- **Disadvantages:**
  - Requires close collaboration with stakeholders, which may be resource-intensive.
  - Less focus on comprehensive documentation, which may pose challenges in maintenance.

3. Spiral Model:

- **Advantages:**
  - Focuses on risk management and can assist in early detection and resolution of problems
  - Focuses on risk management while combining incremental and iterative development.
- **Disadvantages:**
  - Intricate and could necessitate a large amount of risk analysis work.
  - Ineffective management of risk analysis might lead to an increase in project expenses.

4. Iterative Water Model:

- **Advantages**:
  - Allows for revisiting and refining earlier phases, reducing the risk of errors.
  - Lowers the possibility of a major project failing.
  - Suitable for projects where minor adjustments may be required during development.
- **Disadvantages**:
  - Less flexible than Agile methods.
  - Requires careful management to ensure iterations don't cause scope creep.

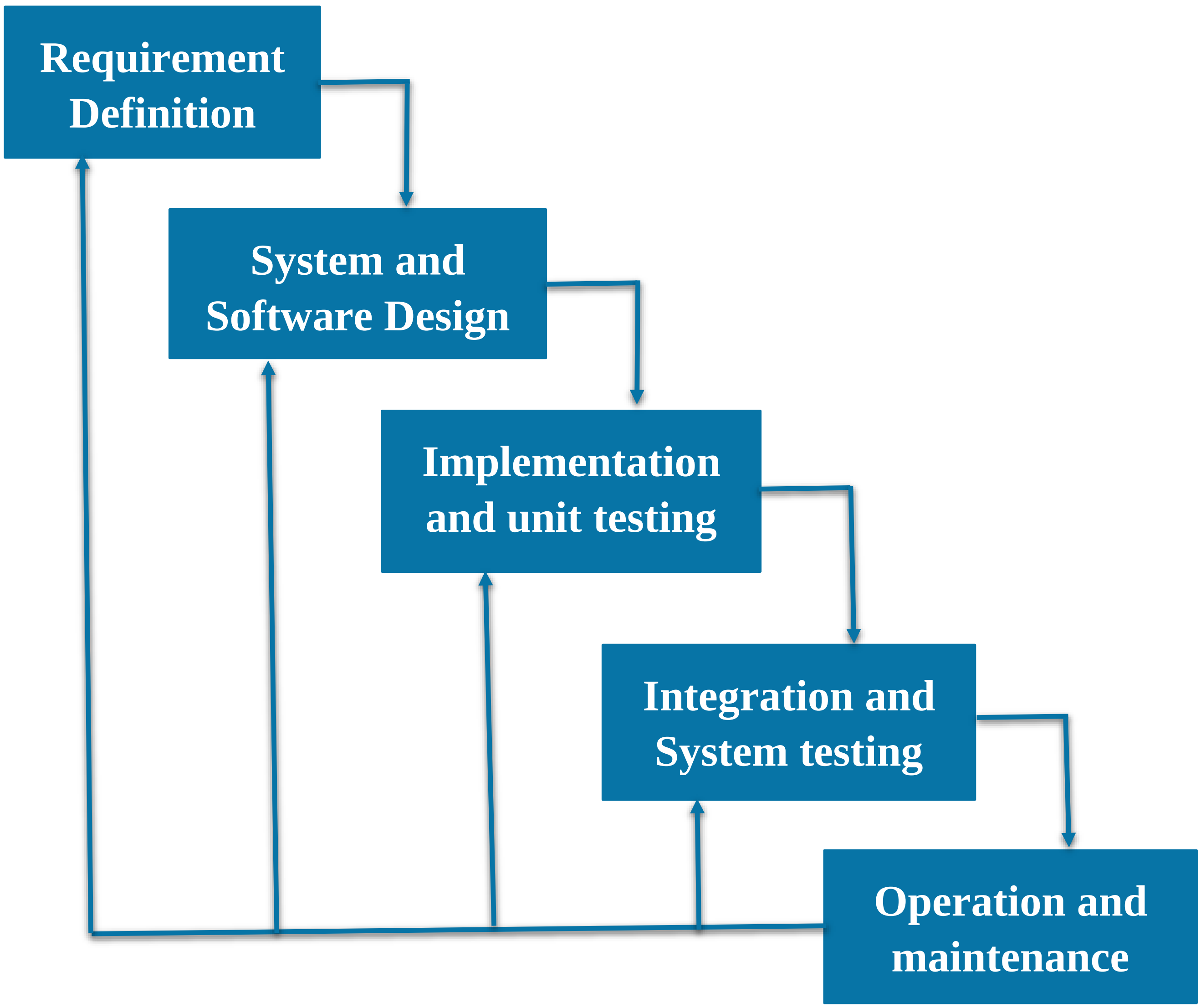
SDLC DIAGRAM

Waterfall Model:

- The Hotel Management System(HMS) follows the waterfall model because:
  - The project has well-defined requirements from the start, like room bookings, billing, and check-ins.
  - The development follows a clear step-by-step process: planning, design, development, testing, and deployment.
  - The requirements are unlikely to change much during the project, which is ideal for the Waterfall approach.

If the system involves periodic feedback or minor adjustments during the design or implementation phases, it could be following a **Modified Waterfall Model**.

Waterfall Model SDLC Diagram



SDLC PHASES FOR HMS

1. **Requirement Analysis**
    - Identify the functional requirements like customer data management, room assignment, billing, and reporting.
    - Gather non-functional requirements such as system performance, scalability, and security.
2. **System Design**
    - Create architectural and database designs.
    - Plan the user interface to ensure a seamless experience for hotel staff.
3. **Implementation**
    - Develop the HMS by coding features like adding customers, calculating costs, and generating bills.
    - Integrate the front-end interface with the back-end database.
4. **Testing**
    - Perform functional testing to ensure features work as expected (e.g., accurate cost calculation).
    - Conduct system testing for performance and compatibility with operating systems and browsers.
5. **Deployment**
    - Install the system on hotel computers and train staff to use the interface.
    - Ensure smooth integration with existing workflows.
6. **Maintenance**
    - Address bugs, update features, and improve system security over time.
    - Monitor performance and adjust for increased user load or data storage.

REQUIREMENTS

FUNCTIONAL REQUIREMENTS

1. **Customer Record Management:**
    - The system must allow users to add, update, and search for customer records efficiently.
    - It should include fields for personal details such as first name, surname, address, date of birth, postcode, mobile number, email, nationality, and gender.
    - The search functionality must retrieve records using criteria like customer name or reference number.
2. **Proof of Identification Storage:**
    - The system must store and display proof of identification such as driving licenses, passports, or student IDs for verification purposes.
    - These records must be securely stored and accessible for audits or guest verification during check-in.
3. **Room Assignment and Management:**
    - Staff must be able to assign rooms to guests based on room type (e.g., single, double, suite) and availability.
    - The system should display current room availability in real-time to prevent double bookings or allocation errors.
4. **Check-in and Check-out Tracking:**
    - The system must record guest check-in and check-out dates to track room usage and plan for housekeeping or maintenance.
    - The recorded dates should update room availability dynamically.
5. **Billing and Financial Calculations:**
    - The system must calculate the guest's stay duration based on check-in and check-out dates.
    - It must apply relevant taxes and display a breakdown of charges, including the subtotal and total cost, to ensure accurate billing.

6. **Bill Generation and Display:**
    - A detailed bill, including charges for room stay, taxes, and any additional services like meal plans, must be generated.
    - The bill must be displayed to the guest at checkout and saved in the database for future reference.
7. **Customer Record Display:**
    - All customer records must be displayed in a table format, providing a clear view of details like booking dates, room numbers, and payment status.
    - The table should support sorting and filtering options for easier navigation.
8. **Data Deletion and Reset:**
    - The system must allow staff to delete outdated or incorrect records to maintain a clean database.
    - A reset option must clear all fields for new data entry without retaining previous information.

NON-FUNCTIONAL REQUIREMENTS

1. **Usability:**

- The system should have an intuitive and user-friendly interface, allowing staff to complete tasks with minimal training.

2. **Performance:**

- The system must handle up to 1000 customer records and support 5 concurrent users without performance degradation.
- Load time of UI Should not take more than 2 seconds
- Login Validation should be done within 3 seconds
- Data in database should be updated within 2 seconds

3. **Security:**

- Customer data must be encrypted to protect sensitive information.

- Role-based access control must ensure only authorized users can access specific features.
- Database should be backed up every hour.
- Under failure, the system should be

4. **Reliability:**

- The system should operate smoothly without crashes, ensuring uninterrupted service during peak hours.
- Database should be backed up every hour.
- Under failure, the system should be able to come back at normal operation under an hour.

USER REQUIREMENTS

1. **Receptionists:**

    - Manage guest check-ins, check-outs, and room assignments.
    - Generate accurate bills, update records, and search for customer details quickly.

2. **Administrators:**

    - Oversee customer records, room availability, and user permissions.
    - Manage financial reports and track system usage for security.

3. **Guests:**

    - Receive accurate bills and benefit from quick check-in/check-out processes.
    - Have personal preferences and details securely stored.

4. **Housekeeping Staff:**

    - Access real-time updates on room statuses for daily tasks.
    - Plan schedules based on check-in and check-out data.

5. **Hotel Managers:**

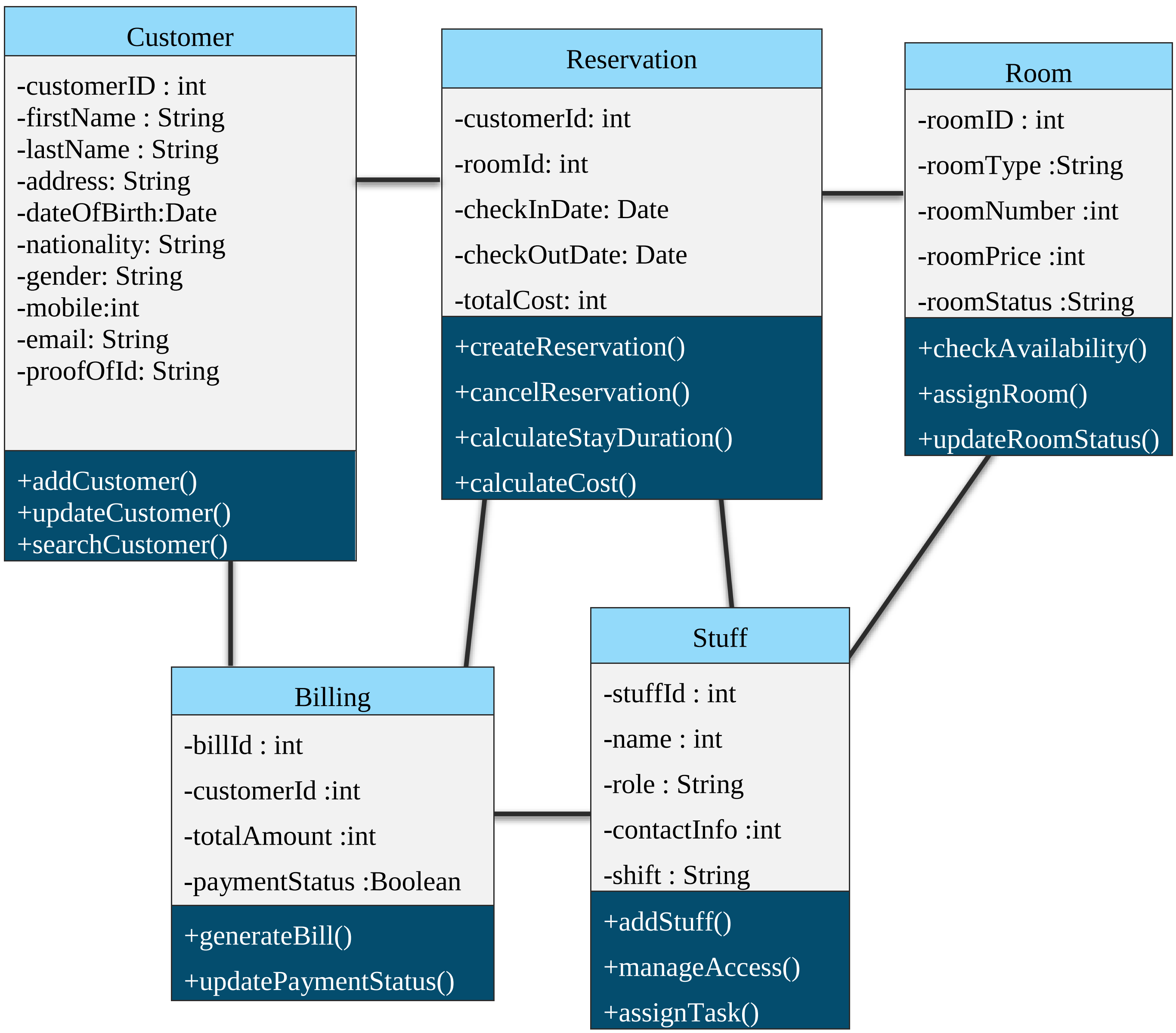
    - Monitor occupancy rates, financial summaries, and staff activities.
    - Approve key updates or changes to bookings and records.

SYSTEM REQUIREMENTS

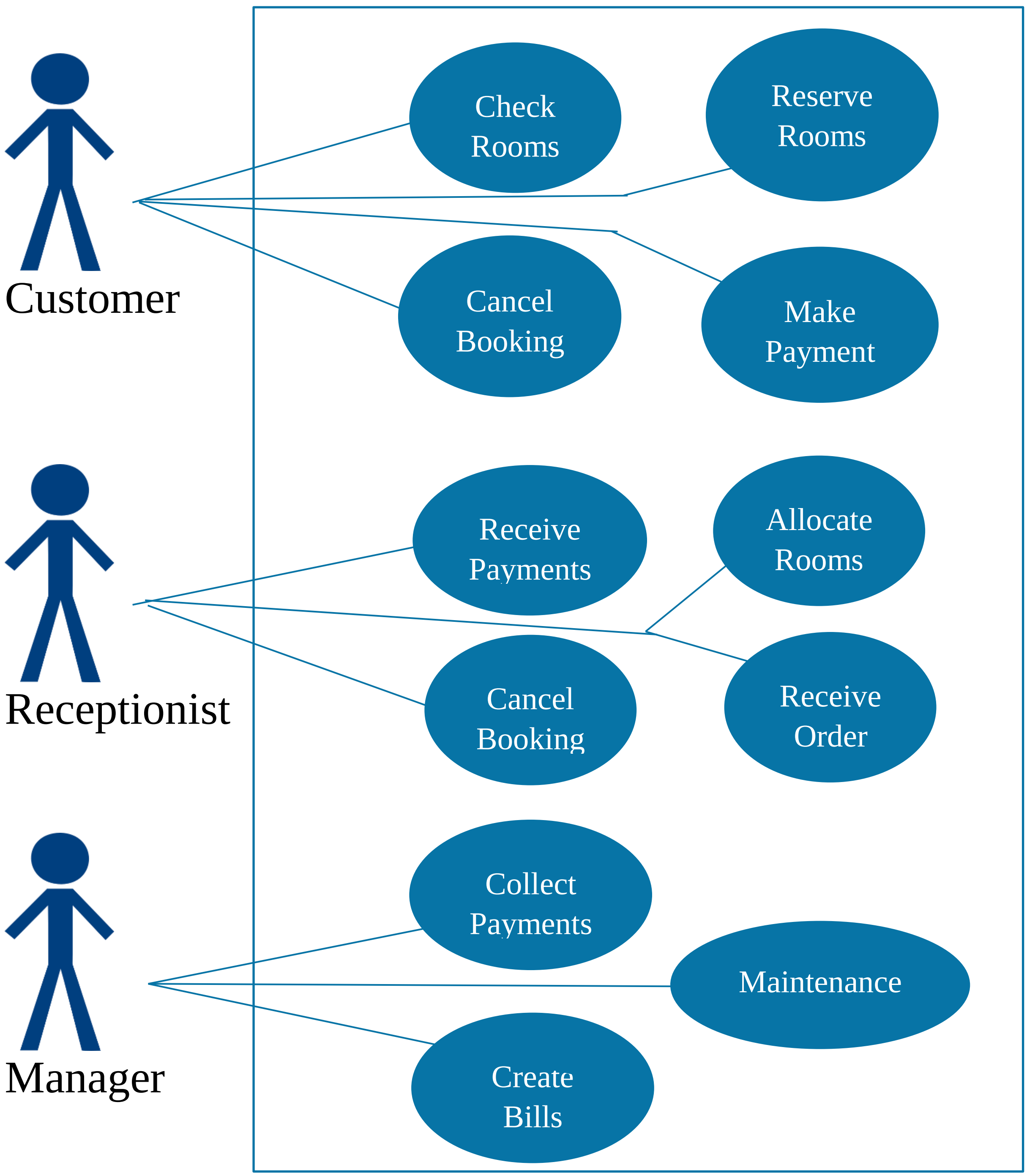
- A dual-core processor and 4 GB of RAM are required for smooth performance and multitasking.
- 500 GB of storage is needed to store customer data and booking details.
- The system should work on Windows 7 or higher, macOS, or Linux.
- A lightweight relational database like SQLite or MS Access will store customer and booking data.
- The system must be compatible with major browsers (Chrome, Firefox, Edge, and Safari).
- It should handle up to 1000 customer records and support 5 concurrent users without performance issues.
- Encryption is necessary to protect sensitive customer data.
- Role-based access control ensures only authorized staff access specific features.
- Backup and recovery systems must be in place to protect data in case of failures.

DIAGRAM

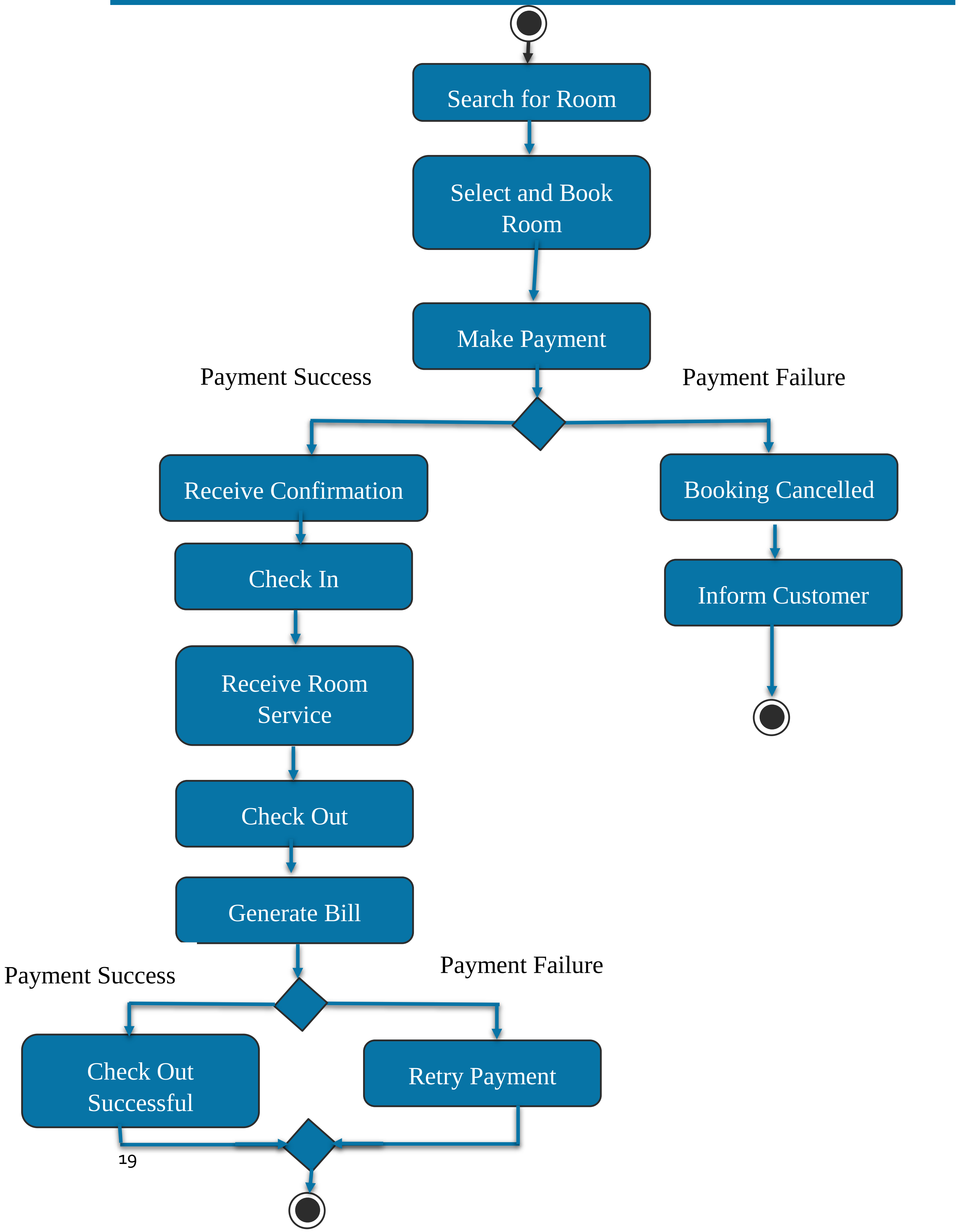
CLASS DIAGRAM



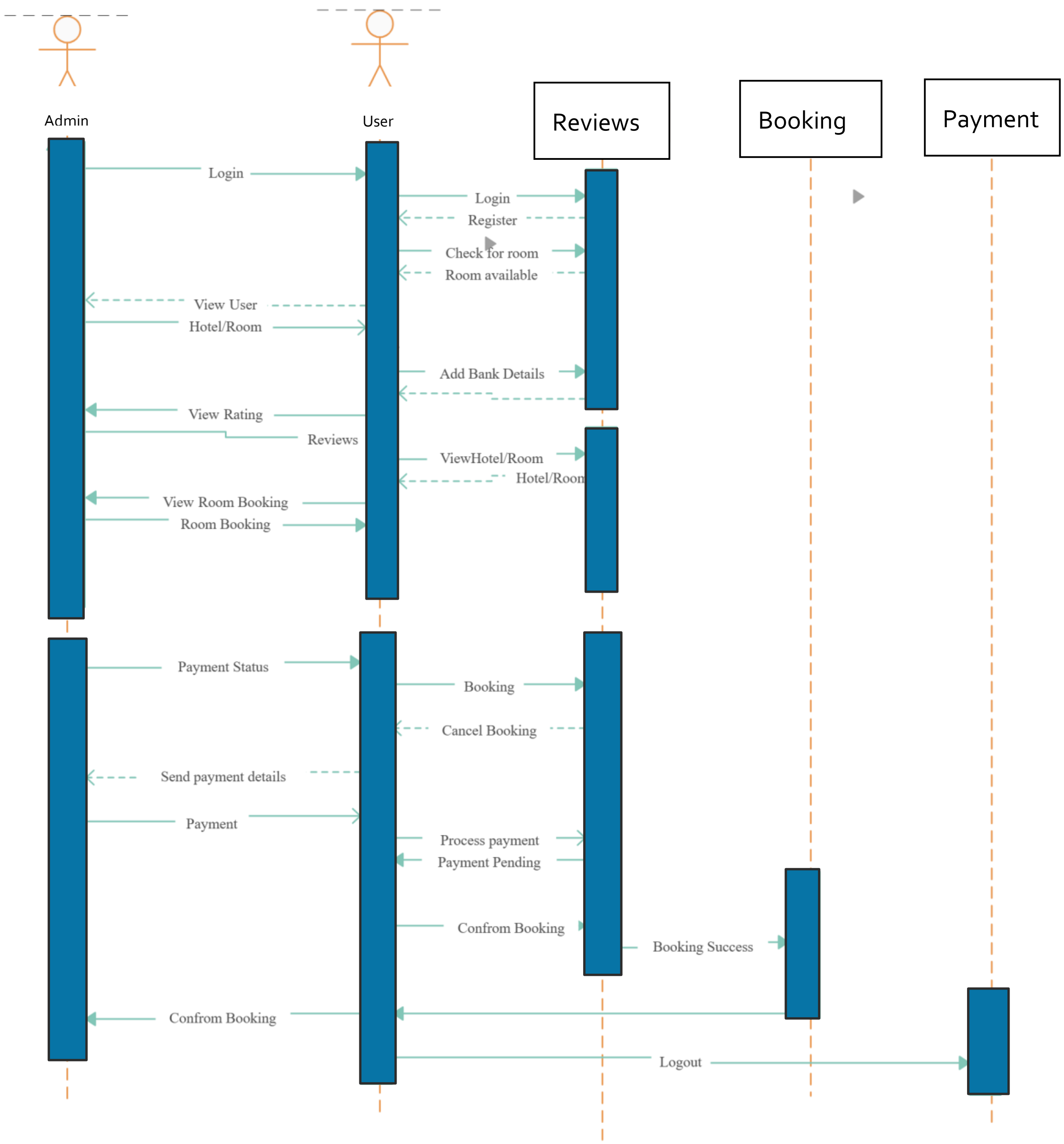
USE CASE DIAGRAM



ACTIVITY DIAGRAM



SEQUENCE DIAGRAM



TESTING

1. Functional Testing

- Verify that all functionalities, such as adding/updating customer records, assigning rooms, and calculating bills, work as expected.
- Check the accuracy of features like total cost calculation, tax inclusion, and duration of stay computation.

2. Unit Testing

- Test individual components like the "Add New" button, room allocation logic, and tax calculation.
- Ensure each module functions correctly in isolation.

3. Integration Testing

- Ensure smooth interaction between modules like the user interface, database, and billing logic.
- Verify data flow between components (e.g., room type selection and cost computation).

4. System Testing

- Test the entire system to ensure it works seamlessly under normal conditions.
- Check compatibility across supported operating systems (Windows, macOS, Linux) and browsers (Chrome, Firefox, Edge, Safari).

5. Performance Testing

- Evaluate system response time when handling up to 1000 customer records or multiple users simultaneously.
- Check the system's performance under heavy loads, such as peak check-in/check-out times.

6. Security Testing

- Ensure sensitive data (e.g., customer identification and payment details) is encrypted.
- Verify role-based access control, ensuring only authorized staff can access specific features.

7. User Acceptance Testing (UAT)

- Allow hotel staff to test the system in a real-world scenario.
- Gather feedback on usability, workflow efficiency, and overall satisfaction.

8. Regression Testing

- Re-test the system after updates or bug fixes to ensure existing functionalities remain unaffected.

9. Backup and Recovery Testing

- Simulate system crashes to verify the backup system can restore data without loss.
- Test recovery processes to ensure continuity of operations.

CONCLUSION

The Hotel Management System (HMS) is designed to automate and optimize critical hotel operations, ensuring efficient workflows, accurate data management, and enhanced guest satisfaction. By implementing this system, hotels can reduce manual effort, minimize errors, and provide a seamless experience for both guests and staff.

With features such as room booking, customer data management, billing, and real-time tracking, the HMS addresses the essential requirements of modern hotel management. The system's scalability and ease of use make it a valuable solution for small to medium-sized hotels aiming to streamline operations and improve service delivery.

By adopting the HMS, hotels can achieve greater operational efficiency, improve customer satisfaction, and maintain a competitive edge in the hospitality industry.

REFERENCE

1. Sommerville, I. (2015). *Software Engineering* (10th Edition). Pearson Education

2. Search Engine:

    - www.google.com

3. Online tools

    - https://creately.com

4. Software Modeling Tool

    - StarUML

5. Class Notes